Lab-2: Object Detection and Segmentation

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In this lab, we will explore real-time object detection using bounding boxes and segmentation, utilizing data recorded via rosbag. We will also cover person keypoint detection.

The tutorial includes two parts:

- **A. Object detection**
  - **A.1 Converting ROS Bag to MP4**
  - **A.2 Case 1: Realtime detection**
  - **A.3 Case 2: MP4-based detection**
  - **A.4 Change the model: compare different algorithms and different datasets**
- **B. Person Keypoint Detection**
- **C. Panoptic Segmentation**

If you are using a virtual machine platform or you don’t have GPU, you can follow these two steps: first, record a bag, and then play it online via Colab.
Please follow the instructions:
https://github.com/hrc-pme/Deep_learning_d415_ROS2/blob/main/Tutorial/maskrcnn.md
This document is also uploaded to GitHub. Tutorial/lab2

A. Object detection

A.1 Converting ROS Bag to MP4(ROSBag2MP4)

If you plan to perform MP4-based detection, please make sure you have followed the steps below to convert your ROS bag into an MP4 video. If you are instead interested in real-time detection, you may skip this section and go directly to **A.2**.

Convert ROSBags to MP4 Video

Edit the configuration file:

```
~/Workspace/Script/config/bags2mp4.yaml
```

Default configuration:

```yaml
jobs:
  - input_paths:                      # Paths to bag folders
      - "/home/hrc/Workspace/Outputs/rosbags/test"
    output_root: "/home/hrc/Workspace/Outputs/mp4/test"       # Output MP4 folder
    topic: "/camera/camera/color/image_raw/compressed"
    # Image topic   If omit → auto-pick if only one; error if multiple
```

> NOTES: If Image topic omit → auto-pick if only one; error if multiple

Convert ROSBags to MP4 Video

```
cd ~/Workspace/Script
python3 bag2mp4.py
```

A.2 Case 1: Real-time Detection (real-time_detection)

In this section, we provide **four predefined object detection modes** for students to experiment with. You can modify the parameter under **#Model task:** to switch between the following options:

- bbox – Bounding Box Detection
- instance – Instance Segmentation
- keypoint – Keypoint Detection
- panoptic – Panoptic Segmentation

All four modes are trained using the COCO dataset. However, in this lab's checkpoint, you are also required to test the Instance Segmentation model trained on the LVIS dataset and compare its performance with the COCO-based model.

In the A.4 section, we will show you where and how to change the model configuration to switch between the COCO and LVIS checkpoints.

Case 1: Real-time Detection

For real-time detection, make sure the camera is active (check D415_tutorial) !!!

Terminal: Start Real-time Mask R-CNN Inference

```
ros2 launch detection realtime_detection.launch.py
```

> NOTES: You'll see a prompt displaying the recommended timer duration to optimize the inference quality: [INFO] [Time_stamp] [realtime_detection_node]: Inference Time: 0000s (avg 0000s). Hint: set timer_period≈0.1

You can change your setting through `realtime_detection.yaml`

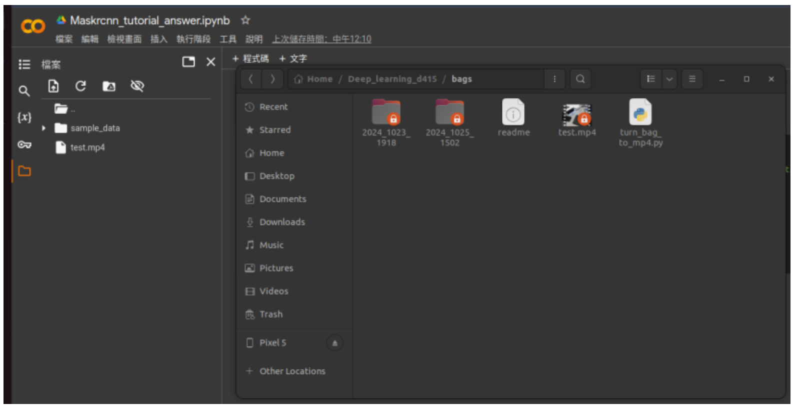
```
file will be located at
/home/hrc/Workspace/ros2_ws/src/detection/config/realtime_detection.yaml

# Confidence threshold
score_thresh: 0.8

# Inference period (s)
timer_period: 0.01

# Model
task: "bbox"   # bbox | instance | keypoint | panoptic

# Computation device
device: "auto"     # cpu | cuda | auto
```

A.3 Case 2: MP4-based Detection([Colab Detection]())

In this section, you will learn how to perform object detection using an MP4 video that was previously converted from a ROS bag. The process will be conducted on Google Colab. (If you don't have GPU it is recommend to use Colab)

A.3.1 Upload

After converting your ROS bag file into an MP4 video, upload the MP4 file to Colab to run the object detection task. Make sure you upload the correct video file in the designated cell.



A.3.2 Initialization

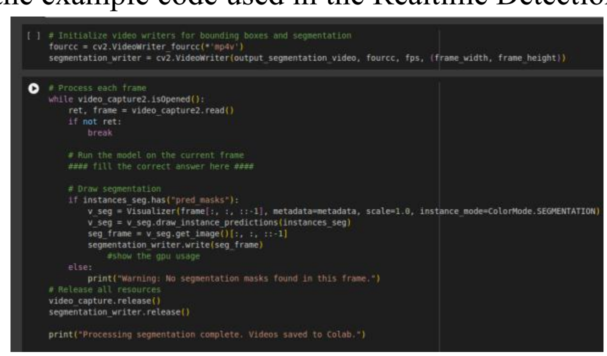
Before starting detection, **verify that the file name is correct**. Check the filename both in the Colab notebook and in your local script (bag2mp4.py) to ensure they match exactly.

A.3.3 Detection (Just Follow the Colab instruction)

Run the detection section in Colab and follow the step-by-step instructions provided in the notebook. This will perform object detection on the uploaded MP4 video.

A.3.4 Segmentation

Next, execute the segmentation section of the notebook. If you are using Colab, you will need to implement how to visualize the segmentation masks yourself. (It's not difficult if you're unsure, you can refer to the example code used in the Realtime Detection section for guidance.)



A.4 Change the model: Compare different algorithms and different datasets

In this section, we will provide instructions on how to change the model configuration. You can use this section to compare different models or experiment with various training hyperparameters to observe their effects on detection performance.

A.4.1 Model Zoo

You can find all available models in the MODEL_ZOO.md. Please take some time to read through the documentation to understand each model's details — including its training configuration, inference time, box AP, mask AP, and other performance metrics.

COCO Object Detection Baselines

Faster R-CNN:

Name	lr sched	train time (s/iter)	inference time (s/im)	train mem (GB)	box AP	model id	download
R50-C4	1x	0.551	0.102	4.8	35.7	137257644	model | metrics
R50-DC5	1x	0.380	0.068	5.0	37.3	137847829	model | metrics
R50-FPN	1x	0.210	0.038	3.0	37.9	137257794	model | metrics
R50-C4	3x	0.543	0.104	4.8	38.4	137849393	model | metrics
R50-DC5	3x	0.378	0.070	5.0	39.0	137849425	model | metrics
R50-FPN	3x	0.209	0.038	3.0	40.2	137849458	model | metrics
R101-C4	3x	0.619	0.139	5.9	41.1	138204752	model | metrics
R101-DC5	3x	0.452	0.086	6.1	40.6	138204841	model | metrics
R101-FPN	3x	0.286	0.051	4.1	42.0	137851257	model | metrics
X101-FPN	3x	0.638	0.098	6.7	43.0	139173657	model | metrics

RetinaNet:

Name	lr sched	train time (s/iter)	inference time (s/im)	train mem (GB)	box AP	model id	download
R50	1x	0.205	0.041	4.1	37.4	190397773	model | metrics
R50	3x	0.205	0.041	4.1	38.7	190397829	model | metrics
R101	3x	0.291	0.054	5.2	40.4	190397697	model | metrics

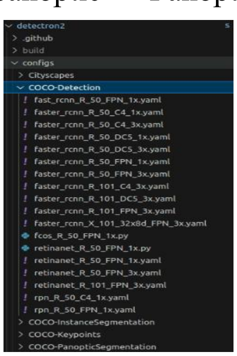
A.4.2 Changing the Model in Realtime Detection

In the file realtime_detection_node.py (~/Workspace/ros2_ws/src/detection/detection/realtime_detection_node.py), you can modify the model by updating the Detectron2 configuration API call. All available model configuration files can be found in: (~/Workspace/detectron2/configs/)

To change the model, simply copy the desired YAML file path and replace it in the corresponding section of the code.

Make sure to match the model type with the correct task mode:

- bbox → Detection
- instance → Instance Segmentation
- keypoint → Keypoint Detection
- panoptic → Panoptic Segmentation




```
# ---- Detectron2 config ----
self.declare_parameter('task', 'instance')  # bbox | instance | keypoint | panoptic
task = self.get_parameter('task').get_parameter_value().string_value

cfg_map = {
    'bbox':     "COCO-Detection/faster_rcnn_R_50_FPN_3x.yaml",
    'instance': "COCO-InstanceSegmentation/mask_rcnn_R_50_FPN_3x.yaml",
    Implement with Codex
    #TODO: change the model to LVIS to be compared with the previous work
    # e.g. 'instance': "LVIS-InstanceSegmentation/mask_rcnn_R_50_FPN_1x.yaml",
    'keypoint': "COCO-Keypoints/keypoint_rcnn_R_50_FPN_3x.yaml",
    'panoptic': "COCO-PanopticSegmentation/panoptic_fpn_R_50_3x.yaml",
}
cfg_name = cfg_map.get(task, cfg_map['instance'])
```

(**REMIND:** LVIS will only be selected for instance → Instance Segmentation)

Now you can select your desired model and start performing the analysis! Try different configurations — including bbox, instance, keypoint, and panoptic modes — and compare the performance of models trained on COCO and LVIS datasets.

Before running the test, make sure to select the correct task in the configuration file: (/home/hrc/Workspace/ros2_ws/src/detection/config/realtime_detection.yaml)

```
# Model
task: "bbox"   # bbox | instance | keypoint | panoptic
```

A.4.3 Changing the Model in MP4-based detection

It follows the same idea as the Realtime detection. You only need to change the model path to the one you want from the Model Zoo. Just make sure that the model you select matches the correct mode — either Detection or Segmentation — depending on your task setting.

Run a pretrained boundingbox detection model with detectron2 framework

```
# Set up Detectron2 configuration
cfg = get_cfg()
cfg.merge_from_file(model_zoo.get_config_file("COCO-Detection/faster_rcnn_R_50_FPN_3x.yaml"))
cfg.MODEL.ROI_HEADS.SCORE_THRESH_TEST = 0.8
cfg.MODEL.WEIGHTS = model_zoo.get_checkpoint_url("COCO-Detection/faster_rcnn_R_50_FPN_3x.yaml")
```

```
# Set up Detectron2 configuration
cfg2 = get_cfg()
cfg2.merge_from_file(model_zoo.get_config_file("COCO-InstanceSegmentation/mask_rcnn_R_50_FPN_3x.yaml"))
cfg2.MODEL.ROI_HEADS.SCORE_THRESH_TEST = 0.8
cfg2.MODEL.WEIGHTS = model_zoo.get_checkpoint_url("COCO-InstanceSegmentation/mask_rcnn_R_50_FPN_3x.yaml")
```

B. Person Keypoint Detection

Use the COCO Person Keypoint Detection Baselines with Keypoint R-CNN to detect and visualize human body keypoints such as the eyes, nose, shoulders, elbows, wrists, hips, knees, and ankles. This task focuses on identifying specific joints or landmarks of each person in the image and drawing skeleton-like connections between them.

C. Panoptic Segmentation

Use the COCO Panoptic Segmentation Baselines to perform comprehensive image understanding that combines both semantic segmentation (classifying every pixel) and instance segmentation (separating individual objects). The model provides a complete scene interpretation, labeling both background areas (like sky, road) and foreground objects (like people, cars).

● Check points

Your lab report (**Lab2_studentID_Name.pdf**) should include the following checkpoints. All **videos must be uploaded to YouTube**, and the **links** should be included in your report.
Do not submit video files directly.

1. Object Detection Comparison (55 points)

Prepare two videos demonstrating:

1. Bounding Box Detection using a RetinaNet or Faster R-CNN model.
2. Instance Segmentation using a Mask R-CNN model.

In your report, analyze the differences between the bounding box and segmentation results, including visual differences, accuracy, and inference speed, and discuss how the algorithms and COCO-based training datasets influence performance.

2. Dataset Comparison (45 points)

Prepare two videos demonstrating:

1. Instance Segmentation using Mask R-CNN (COCO dataset).
2. Instance Segmentation using Mask R-CNN (LVIS baseline).

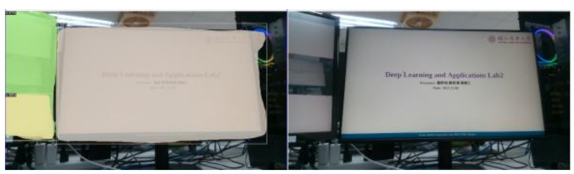
In your report, analyze the differences between COCO and LVIS models, focusing on class diversity, segmentation quality, and detection performance.

3. Bouns(Ubuntu User): Real-time Detection (5 points)

If you are using an Ubuntu system, attempt to perform real-time inference using a **keypoint** model and record a ROS bag. Use the command below to verify that the recorded topics are correct:

```
$ ros2 bag info your_recorded_file.bag
```

Include a **screenshot of the ROS bag info and a frame of the detection result** in your report.
Shortly describe what you have done.



(this is an example not the right answer)

```
hrc@hrc-MS-7D98:~/Workspace/Outputs/rosbags/test$ ros2 bag info 0011

Files:             0011_0.db3
Bag size:          80.5 MiB
Storage id:        sqlite3
Duration:          6.963736647s
Start:             Nov  3 2025 21:35:23.664550745 (1762176923.664550745)
End:               Nov  3 2025 21:35:30.628287392 (1762176930.628287392)
Messages:          253
Topic information: Topic: /camera/camera/color/image_raw/compressed | Type: sensor_msgs/msg/CompressedImage | Count: 23 | Serialization Format: cdr
                   Topic: /camera/camera/depth/image_rect_raw/compressedDepth | Type: sensor_msgs/msg/CompressedImage | Count: 147 | Serialization Format: cdr
                   Topic: /camera/color/image_detection/compressed | Type: sensor_msgs/msg/CompressedImage | Count: 82 | Serialization Format: cdr
                   Topic: /tf_static | Type: tf2_msgs/msg/TFMessage | Count: 1 | Serialization Format: cdr
```

4. Bouns(WSL User): MP4-based Detection_Segmentation (5 points)

Try to complete the TODO part in the segmentation section of the Colab notebook. For example, displaying segmentation masks on the detected objects. Include a **screenshot of the Segmentation Inference and a frame of the detection result** in your report. **Shortly describe what you have done.**